

Library Management System - JavaScript Project Task

Project Objective

Develop a Library Management System using HTML, CSS, and JavaScript. The application should allow users to add, search, issue, return, and manage books efficiently.

Required Features

1. Add Book
2. Display All Books
3. Search Book by Name, Author, or Category
4. Issue Book
5. Return Book
6. Delete Book
7. Book Availability Status
8. Display Total Books Count

Book Information

- Book ID
- Book Name
- Author Name
- Category
- Quantity

Book Object Structure

```
{ id: 1, bookName: 'JavaScript Basics', author: 'Raj Patel', category: 'Programming', quantity: 5 }
```

Availability Rules

Quantity > 0 → Available
Quantity = 0 → Out of Stock

Functions To Implement

```
addBook()  
displayBooks()  
searchBook()  
issueBook()  
returnBook()  
deleteBook()  
getBookStatus()
```

JavaScript Topics That Must Be Used

DOM Manipulation
Variables (let, const)
Arrays
Objects
Functions
Conditions (if-else)
Loops (for-of)
Array Methods (push, filter, find, sort)
Template Literals

Suggested User Interface

Book Name Input
Author Input
Category Input
Quantity Input
Add Book Button

Search Box

Table Columns:

ID | Book Name | Author | Category | Quantity | Status | Actions

Evaluation Checklist

- Add Book Working
- Search Working
- Issue Book Working
- Return Book Working
- Delete Book Working
- Availability Status Working
- Total Books Count Displayed
- Clean User Interface
- Proper Use of Arrays and Objects